

Nifty 50 Behaviour Analysis

A detailed technical report — the objective, the data, the buy-before / sell-after methodology, the win-rate mathematics, current results, the data-quality work, the technology stack, and the honest limitations of the study.

Detailed Project Report

Universe: the current Nifty 50 (verified 1:1 with NSE)

Data: Yahoo Finance · NSE India · Screener.in

Status: local build · for study and education, not investment advice

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1 Executive summary

This project studies **how the 50 Nifty stocks move around their corporate events** — quarterly results, board meetings, corporate actions and material announcements — to test one disciplined idea: **buy a few days before a known, scheduled event and sell a few days after**. For every stock and event type it brute-force tests **64 timing rules** against the full event history, reports the best rule and its **win rate**, and serves everything through a self-built web dashboard.

50

stocks (exact Nifty 50)

68,589

unique event-dates

106,034

feed rows analysed

50/50

fundamentals + prices

The honest headline: past win-rates around events are real but **in-sample and modest** — a description of history, not a proven money machine. The project's value is the **transparent, reproducible framework** and the **clean, verified, fully-historical Nifty 50 dataset** behind it.

2 Objective & core idea

2.1 The question

Do Nifty 50 stocks show a **repeatable price pattern** around their corporate events — and if so, what is the best number of days to enter before and exit after each event type?

2.2 The hypothesis: anticipation + drift

Because event dates are published in advance, markets partly **anticipate** them (price drifts up into a known results date) and partly **digest** them afterwards (price keeps drifting for days after the news). The strategy sits inside both windows.

BUY BEFORE

Catch the anticipation

Enter **N days before** to sit inside the run-up toward a date the market already knows.

SELL AFTER

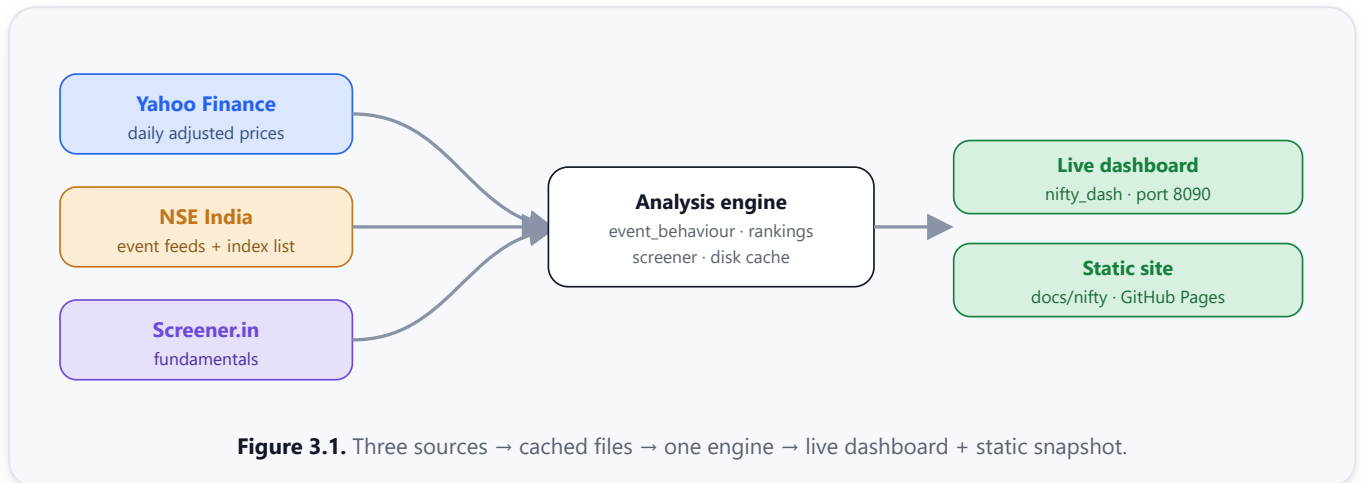
Catch the drift

Exit **M days after** so the post-event drift plays out instead of selling into the first spike.

Key framing. We never forecast *whether* an event happens — the date is read off the regulatory calendar. We only study the *price behaviour* around that fixed date.

3 Data sources & pipeline

3.1 Three external sources, one engine



DATA	SOURCE	FROM WHERE (ENDPOINT / FILE)	CACHED AS
Daily prices	Yahoo Finance	query1.finance.yahoo.com/v8/finance/chart/<SYM>.NS	processed/price_cache/
NIFTY index	Yahoo Finance	.../chart/^NSEI	price_cache/_NSEI.csv
Event feeds	NSE India	nseindia.com/api/corporates-* + NSE export (.xlsx)	<SYM>/*.csv
Fundamentals	Screener.in	screener.in/company/<SYM>/ + schedules API	pn1/ balance_sheet/ ...
Nifty 50 list	NSE India	equity-stockIndices?index=NIFTY 50 · Company Directory	download_feeds.py

3.2 Caching & offline operation

Every source is snapshotted to disk on first fetch. Prices use **adjusted close** (corrected for splits/bonuses). Once downloaded, the entire system — dashboard included — runs **fully offline**.

4 The Nifty 50 universe

4.1 The sync

The stock list was re-synced to the **exact current Nifty 50** from the official NSE Company Directory. The old list had drifted to 52 symbols (stale members never removed through index reshuffles).

+4 added

ETERNAL (ex-Zomato) · INDIGO · MAXHEALTH · TMPV
(Tata Motors PV)

–6 removed

BPCL · BRITANNIA · HEROMOTOCO · INDUSINDBK · LTIM
· TATAMOTORS

4.2 Verification

Result: dashboard universe == NSE Company Directory == NSE MarketWatch export — **all three identical, zero discrepancies.** Every company shown is a current Nifty 50 constituent (the full list is in Appendix A).

5 The dataset

After refreshing from the NSE export and merging the full historical archive, the 50 stocks carry:

FEED	UNIQUE DATES	ANCHOR COLUMN	WHAT IT IS
Announcements	63,021	sort_date	Press releases, acquisitions, ratings, outcomes.
Financial results	3,158	broadCastDate	Quarterly/annual results, back to ~2007.
Board meetings	1,553	bm_date	Board-meeting dates, backfilled to 2006.
Corporate actions	857	exDate	Dividend / bonus / split ex-dates.
Total	68,589		across 106,034 raw feed rows; fundamentals + prices for 50/50.

6 Methodology — the buy-before / sell-after trade

6.1 The two-leg trade

For an event pinned to trading day **T**: **buy N trading days before** at the close, hold through the event, **sell M trading days after** at the close. You are never flat during the event.

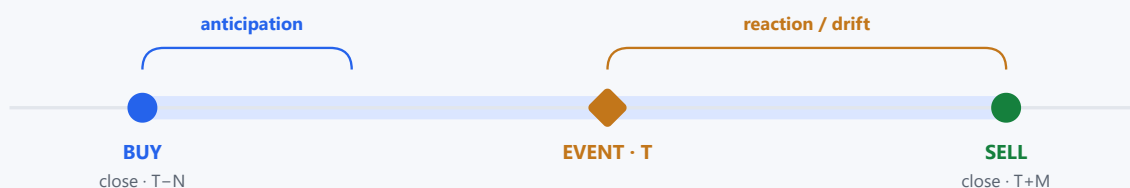


Figure 6.1. One buy, one sell, held through the event.

6.2 Anchoring & the 64-rule grid

T = the first trading day on/after the event date. We then test every combination of $N = 1 \dots 8$ days before and $M = 1 \dots 8$ days after — an $8 \times 8 = 64$ -rule grid. Days are **trading days** (weekends/holidays skipped).

```
entry = close[T - N] // N trading days before
exit  = close[T + M] // M trading days after
return% = ( exit / entry - 1 ) × 100 → WIN if > 0 else LOSS
```

6.3 Worked calendar — “buy 3 before, sell 2 after”

Results broadcast on **Thursday 24 July**. Note the sell lands on **Monday** — trading days skip the weekend.

OFFSET	DATE	WEEKDAY	ACTION
T-3	21 Jul	Mon	BUY at close
T-2 ... T-1	22–23 Jul	Tue–Wed	hold
T	24 Jul	Thu	EVENT — hold through
T+1	25 Jul	Fri	hold
–	26–27 Jul	Sat–Sun	market closed — not counted
T+2	28 Jul	Mon	SELL at close

7 The win-rate mathematics

7.1 The formula

Pool all past events of a type; for each of the 64 cells, the win rate is the share that finished green:

$$\text{win rate}(N,M) = (\text{\#events with return\%} > 0) \div (\text{total events}) \times 100$$

7.2 Worked example — 6 past results, rule “buy 2 / sell 1”

RESULT #	ENTRY	EXIT	RETURN %	OUTCOME
1	1,000	1,032	+3.2	WIN
2	1,120	1,109	-1.0	LOSS
3	980	1,005	+2.6	WIN
4	1,210	1,228	+1.5	WIN
5	1,150	1,142	-0.7	LOSS
6	1,300	1,339	+3.0	WIN

$$\text{wins} = 4 / 6 \rightarrow \quad \cdot \quad \text{avg return} = (3.2 - 1.0 + 2.6 + 1.5 - 0.7 + 3.0) / 6 =$$

7.3 Choosing the one displayed rule

- 1 **Require enough events** — keep cells backed by ~5+ events.
- 2 **Keep the profitable cells** — average return must be positive.
- 3 **Rank by win rate, then average return** — the top cell becomes the displayed rule.

8 How we “predict” all four events

We read the date off the regulatory calendar. The four events split by **how knowable the date is**:



Figure 8.1. Three events have a date fixed in advance; announcements do not.

EVENT	ANCHOR (T)	HOW THE DATE IS KNOWN IN ADVANCE	BUY-BEFORE REAL?
Quarterly Results	broadCastDate	SEBI LODR Reg 29 — results board meeting intimated ≥ 5 clear days ahead.	Yes
Board Meeting	bm_date	Same Reg 29 intimation announces the meeting date ~5 days ahead.	Yes
Corporate Action	exDate	NSE broadcasts the record/ex-date days-to-weeks beforehand.	Yes
Announcement	sort_date	Unscheduled — press releases, acquisitions, ratings arrive without warning.	No*

***Announcements.** You cannot know the date ahead, so the buy-before leg is *not* predictive — its win rate should be read only as **post-disclosure drift** and treated with extra caution.

Corporate actions use adjusted prices. The ex-date drop is removed, so the win rate is **residual total-return drift** — not dividend capture.

9 Current results — rankings leaders

Top-3 stocks by win rate for each event type, from the live system. Read every win rate **next to its event count (n)** — a high rate on a tiny n is weak evidence.

EVENT	#	STOCK	RULE	WIN %	AVG %	N
Quarterly Results	1	TMPV	buy 1 · sell 2	86	+6.14	7
	2	APOLLOHOSP	buy 2 · sell 4	74	+3.02	74
	3	NESTLEIND	buy 6 · sell 7	71	+0.05	7
Board Meeting	1	HINDALCO	buy 1 · sell 4	92	+3.55	12
	2	EICHERMOT	buy 3 · sell 3	86	+4.07	7
	3	M&M	buy 8 · sell 1	84	+4.54	19
Corporate Action	1	SBILIFE	buy 2 · sell 8	100	+3.79	8
	2	LT	buy 8 · sell 5	100	+6.74	1
	3	JIOFIN	buy 8 · sell 5	100	+3.60	1
Announcement	1	MAXHEALTH	buy 6 · sell 8	73	+4.90	74
	2	NESTLEIND	buy 3 · sell 7	71	+1.18	35
	3	APOLLOHOSP	buy 8 · sell 7	66	+2.71	226

Read the sample size. SBILIFE 100% (n=8) is meaningful; LT and JIOFIN 100% (n=1) are single events and essentially noise. APOLLOHOSP announcements (66%, **n=226**) is the most statistically robust row here.

10 Data-quality work

- A Merged the full history.** The recent NSE export had truncated the financial-results history (e.g. Reliance to 4 quarters). Merging with the original archive **recovered +2,860 unique event-dates** — restoring 71–83 quarters per stock.
- B Fixed board-meeting duplication.** The merge had doubled board meetings; de-duplicating to one row per date **removed 617 duplicate rows** ($2.22\times \rightarrow 1.07\times$).
- C Backfilled board meetings 4×.** NSE's feed holds only ~2 recent years; pulling historical "Outcome of Board Meeting" records from the announcements archive (to 2006) lifted the average from **~9 to ~38 meetings per stock**.

Why win-rates dropped slightly after the merge (e.g. an announcement rule 61% $\rightarrow$ 57%): more events regress a fluke small-sample figure toward an honest one — lower but more trustworthy.

11 Technology stack

11.1 Methods

Event-study window · exhaustive 64-rule grid search · descriptive statistics · trading-day indexing · market adjustment vs NIFTY · train/test + placebo validation · fingerprinted disk cache. **No machine learning** (see §14).

11.2 Tools — what, from where, used where

TOOL	PURPOSE	FROM WHERE	USED IN
Python 3	Backend language.	python.org	all scripts
pandas	Tables, CSV, dates.	PyPI	~all scripts
numpy	Numeric math.	PyPI	event/optimal scripts
openpyxl	Excel export.	PyPI	stock_server.py
urllib · ssl · json	Fetch & parse.	Python stdlib	fetchers + server
http.server	Web server (no framework).	Python stdlib	stock_server.py
HTML · CSS · JS · SVG	Dashboard UI & charts.	hand-written	stock_server.py
Headless Chrome	HTML $\rightarrow$ PDF.	Google Chrome	these reports
Git · GitHub Pages	Version control & static host.	github.com	docs/nifty/

12 The dashboard & its tabs

TAB	SHOWS	LOGIC BEHIND IT
Fundamentals	P&L, balance sheet, ratios, quarterly	Reads local Screener CSVs; offline Excel export.
Behaviour	The 64-cell grid + best rule + past-event paths	The core engine, per stock.
Rankings	All 50 ranked by win rate, per event type	Runs the engine across all 50; cached.
Screener	Filter by ROCE / growth / debt / P/E	Latest annual fundamentals per stock.
Compare	2–3 stocks side by side	Fundamentals + behaviour together.
Event feeds	Announcements / Board / Corp Actions / Results	The raw per-stock CSVs, newest first.

13 Performance & caching

The feeds grew large, so computing rankings live took ~40 s (and stalled the browser under load). The fix: rankings & screener now **persist to a disk cache keyed by a data fingerprint** — instant loads, auto-refresh only when the data changes; the cache also warms at startup.

ENDPOINT	BEFORE	AFTER
Rankings	~190 s (hung)	0.09 s
Screener	~2.5 s	0.003 s
Behaviour / feeds	—	< 0.7 s

14 On machine learning

The live system uses **no machine learning** — only grid search and statistics, deliberately, for full transparency. A standalone experiment (`ml_model.py`, scikit-learn) tested whether a model could predict the results trade using price + fundamental features. Its honest conclusion: with only ~48 training examples per stock, extra features **hurt** (curse of dimensionality) and the model **could not reliably beat the trivial baseline** ("always predict WIN" $\approx 62\%$). ML is therefore not part of the product.

15 Limitations & findings

In-sample bias. The best (N,M) is chosen from 64 tries on the same history it is scored on, so win rates flatter. They describe the past, not forward probabilities.

Small samples. 100% on 1–3 events $\ll$ 60% on 30. Always read the win rate next to n.

Event overlap. Board meetings and results share dates where the meeting *was* the results approval.

Corporate actions $\neq$ dividend capture (adjusted-close, \$8).

Overall finding. Under honest, market-adjusted testing, **no robust, reliably tradeable edge survives out-of-sample** on the Nifty 50. The project's worth is the transparent framework and the clean, verified, fully-historical dataset — a solid base for further, more careful research.

16 Scripts reference

SCRIPT	PURPOSE
event_behaviour.py	The engine — 64-rule grid of returns & win rates.
stock_server.py	Dashboard app — payloads, feeds, rankings, screener, compare, caching.
nifty_dash.py	Nifty-50 front end (port 8090) — rebrands + warms caches.
download_feeds.py	Nifty 50 list + NSE feed downloader.
import_new_data.py	Splits the NSE Excel export into per-stock CSVs.
merge_event_history.py	Unions the export with the archive (recovers full history).
enrich_board_meetings.py	Backfills board meetings from the announcements archive.
build_static_nifty.py	Bakes the dashboard into a static GitHub Pages site.
fund_loader.py	Loads the Screener fundamentals folders.

A Appendix — the 50 constituents

Verified 1:1 against the NSE Company Directory and MarketWatch export.

ADANIENT	HDFCBANK	ONGC
ADANIPTS	HDFCLIFE	POWERGRID
APOLLOHOSP	HINDALCO	RELIANCE
ASIANPAINT	HINDUNILVR	SBILIFE
AXISBANK	ICICIBANK	SBIN
BAJAJ-AUTO	INDIGO	SHRIRAMFIN
BAJAJFINSV	INFY	SUNPHARMA
BAJFINANCE	ITC	TATACONSUM
BEL	JIOFIN	TATASTEEL
BHARTIARTL	JSWSTEEL	TCS
CIPLA	KOTAKBANK	TECHM
COALINDIA	LT	TITAN
DRREDDY	M&M	TMPV
EICHERMOT	MARUTI	TRENT
ETERNAL	MAXHEALTH	ULTRACEMCO
GRASIM	NESTLEIND	WIPRO
HCLTECH	NTPC	

Nifty 50 Behaviour Analysis · Detailed Report · generated from the live system · win rate = share of past events where the chosen rule closed positive · numbers are in-sample · for study and education, not investment advice.